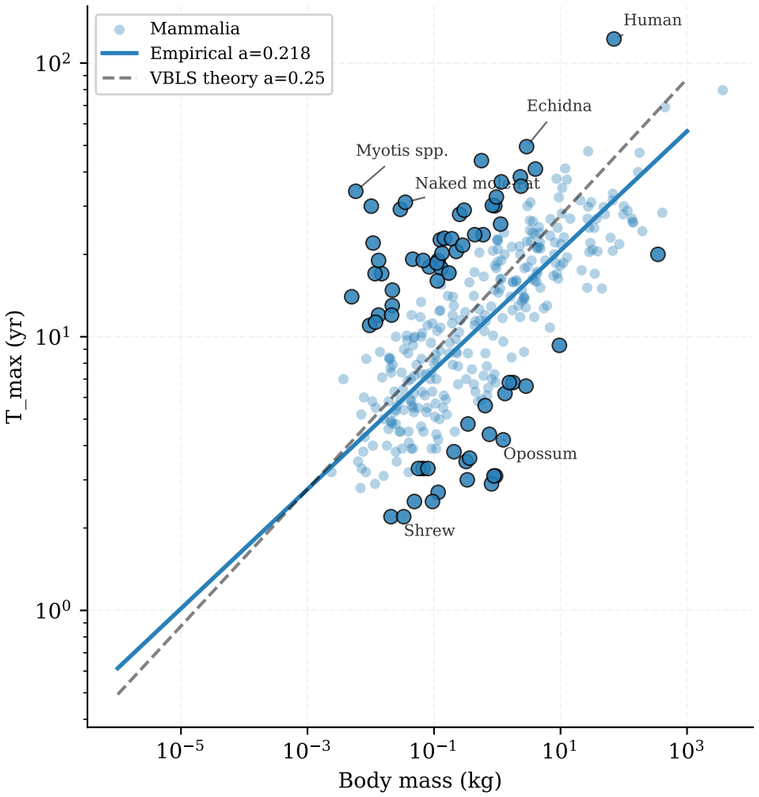
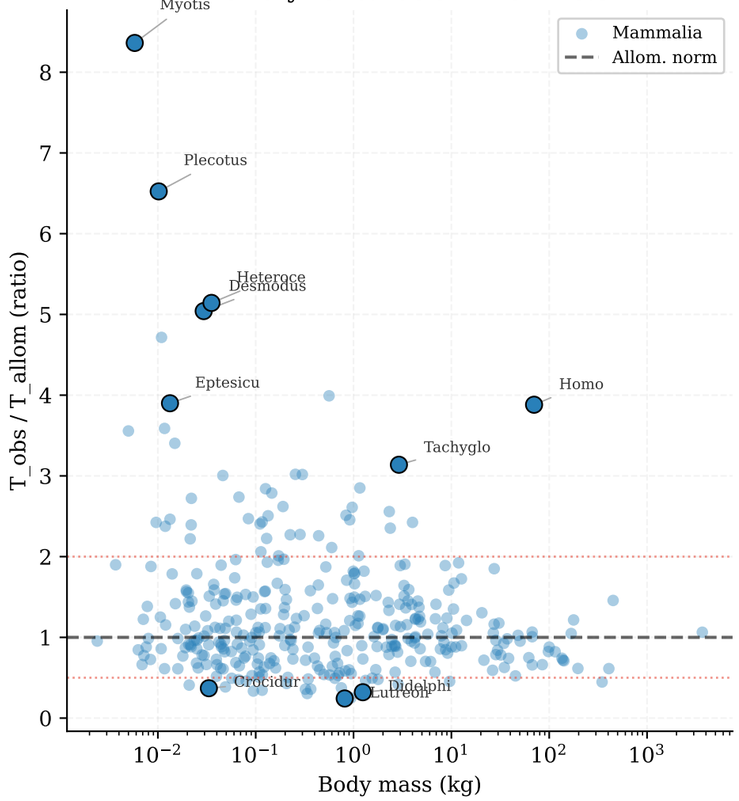
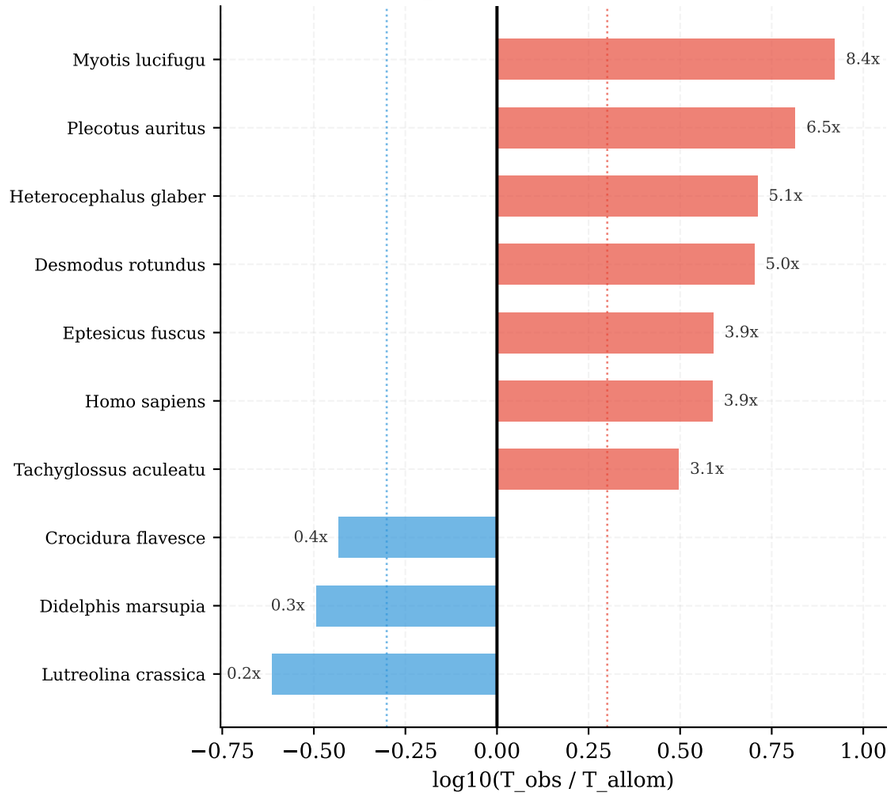
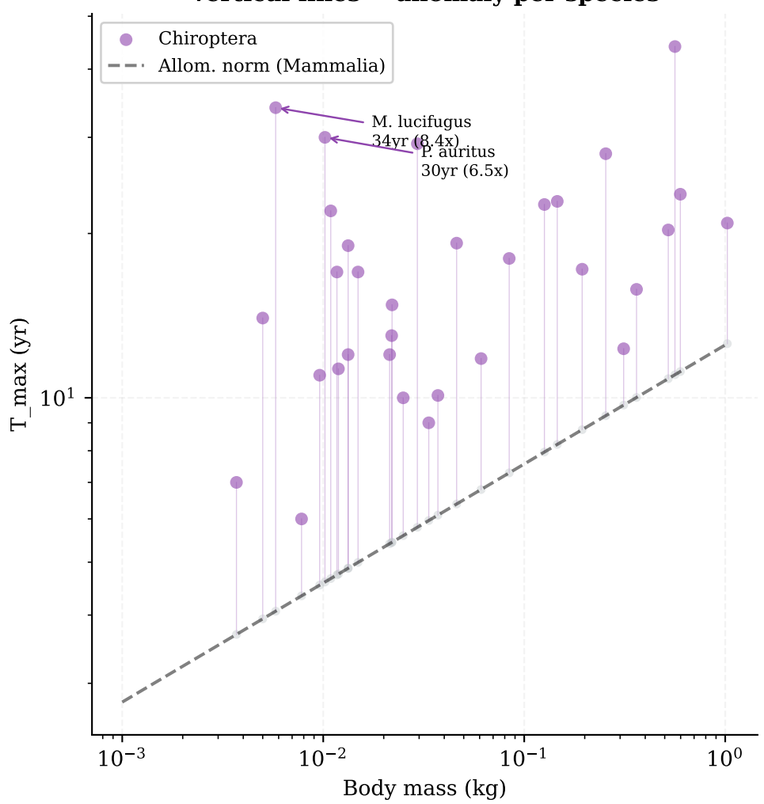






D. Anomaly typology in terms of VBLS
Direction of each shift follows from (S); species are examples

Type	Ratio	Parameter of (S)	Mechanism (examples)
Damage protection	>2	sigma0_eff down	Fewer damage events per unit of dissipation (bats: Myotis)
Slow reserve wear	>2	nu_eff down	Damage kept low, reserve wears slowly (naked mole-rat)
Hibernation	>2	nu0_eff down	Effective < chronological lifespan (Plecotus, Eptesicus)
Lowered body T	>2	nu0 down (T)	Slower degradation reactions (echidna, ~32 C)
Metabolism vs norm	<0.5 / >2	sigma0*F_diss up/down	More / fewer damage events per unit time (no example here)
r-strategist	<0.5	rho down	Lowered restoration of the reserve (opossums)



F. Summary: assigned VBLS parameters

```
Parameters assigned to the deviations
(interpretation; checked for compatibility only)

^ ANOMALOUSLY LONG-LIVED:

Chiroptera (Myotis, x8.4):
  sigma0_eff down: long life at high metabolism;
  Myotis mitochondria release 2-3x less H2O2
  per O2 than shrew and mouse (Brunet-Rossinni 2004)

Heterocephalus glaber (x5.1):
  nu_eff down + sigma0*F_diss down: low metabolism,
  DNA repair genes up vs mouse (MacRae 2015)

Homo sapiens (x3.9):
  sigma0 down + Phi_cap up: medicine prevents
  damage and repairs from outside (cultural)

Tachyglossus aculeatus (x3.1):
  nu0 down + sigma0*F_diss down: body T ~32 C,
  BMR 25-40% of eutherian (Nicol 2017)

v ANOMALOUSLY SHORT-LIVED:

Crocidura flavescens (x0.37):
  not established: metabolism 7.5 W/kg (AnAge),
  as in long-lived bats

Didelphis (x0.32):
  rho down: r-strategist (disposable soma)

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No contradiction with (S) among the ten.
Short-lived records: mostly small captive samples.
Mammalia only: one class, one baseline.
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